

Manifold-Informed Architecture Benchmark

DIGITS: 10 classes, 64D input

Generated: 2026-04-14 20:54:31 | Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD

waverider @ 4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07) | 2026-04-14 20:51:52
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Best: Standard (128→64) test acc = 0.9787 ± 0.0051

Manifold: 64D → 14D (78.1% noise dimensions)

Experimental Setup

Parameter	Value
Dataset	DIGITS
Input dimensionality	64D
Classes	10
Intrinsic dim (d)	14
Variance threshold (τ)	0.9
Epochs	50
Trials	3
Batch size	64
Learning rate	0.001
k neighbors (local PCA)	30

Manifold Discovery

Local PCA over the training set, $k = 30$ neighbors. Intrinsic dimensionality d is taken as the maximum per-class maximum at $\tau = 0.9$, clamped to $n_{\text{classes}} = 10$.

τ	Mean d	Std	Min	Max	Noise (%)
0.95	14.3	1.7	8	18	77.6
0.90	10.9	1.5	5	14	83.0
0.85	8.8	1.4	4	11	86.3
0.80	7.3	1.3	3	10	88.6

Per-Class Intrinsic Dimensionality

Class	Mean d	Std	Min	Max
Digit 0	12.6	0.6	11	14
Digit 8	12.5	1.0	7	13
Digit 9	11.8	0.7	10	13
Digit 5	11.6	0.9	10	13
Digit 3	11.5	0.7	10	13
Digit 6	10.7	1.3	4	12
Digit 7	10.6	1.1	6	12
Digit 2	10.2	0.9	8	11
Digit 4	9.4	1.2	5	11
Digit 1	8.8	1.7	5	11

Architecture Comparison

Architecture	Params	Test Acc ($\mu \pm \sigma$)	Loss	Acc/Kparam
Euclidean KNN (k=7)	0	0.9733 ± 0.0054	—	—
ManifoldModel ($\tau=0.9$)	0	0.9727 ± 0.0054	—	—
	17,226	0.9787 ± 0.0051	0.0825	0.0568
Standard (128→64) ★				
Wide Manifold ($d \rightarrow d+1 \rightarrow d$, $d=14$)	1,509	0.9521 ± 0.0061	0.1780	0.6310
Manifold ($2d \rightarrow d$, $d=14$)	2,376	0.9668 ± 0.0079	0.1227	0.4069
PCA→14D + MLP ($2d \rightarrow d$)	976	0.9375 ± 0.0116	0.2006	0.9605
Intrinsic Dim (PCA→14D→output)	360	0.9178 ± 0.0108	0.2837	2.5495

★ = best performing architecture

Key Findings

- **Best architecture:** Standard (128→64) — test accuracy 0.9787 ± 0.0051
- **Manifold compression:** 64D $\rightarrow$ 14D (78.1% of ambient dimensions carry no structure)

Result Figure

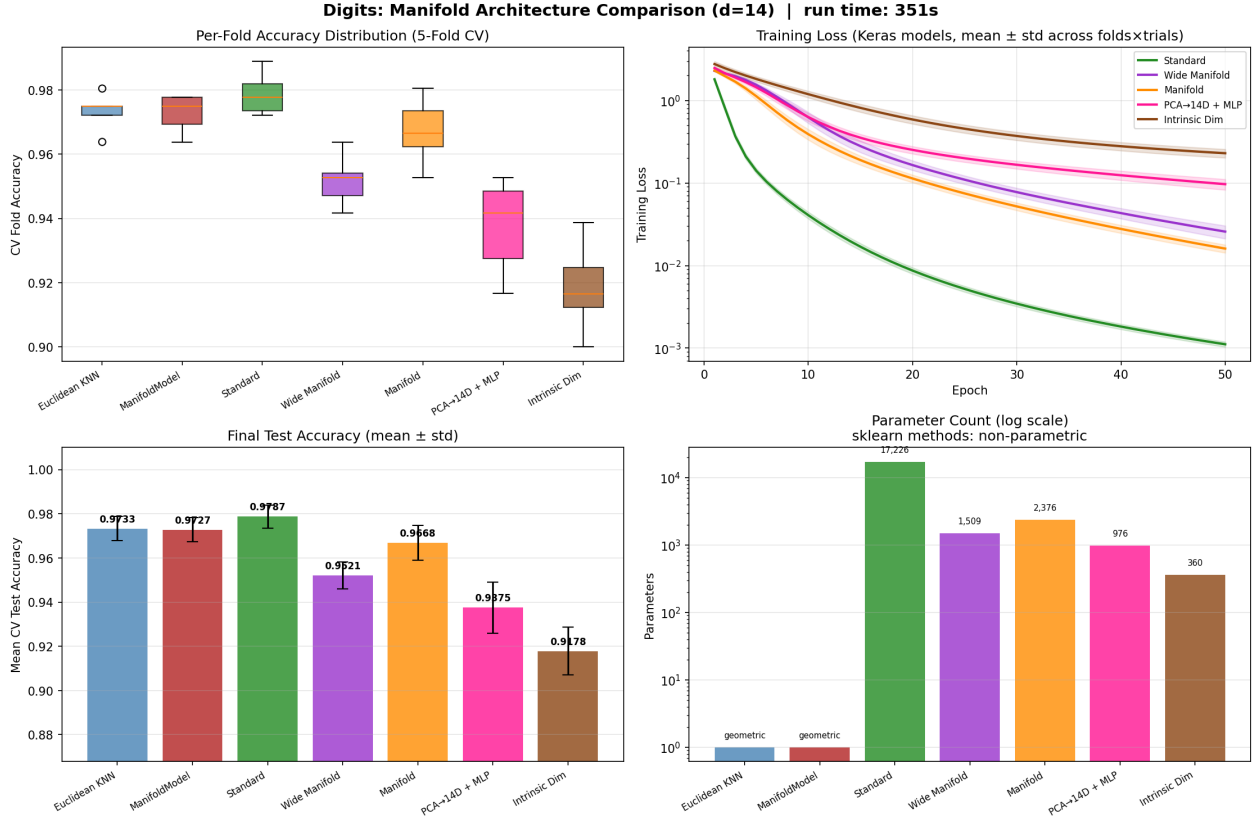


Figure 1: DIGITS manifold-informed vs. standard architecture comparison. $d = 14$, 50 epochs, 3 trials.

Provenance

Item	Value
Repository	Flux-Frontiers/waverider
Commit	4b8002e (-abbrev-re 4b8002ee9a2e3d56a219d7dab695a80b8efd1e07)
Commit date	2026-04-14 20:51:52 -0400
Commit message	add: cifar10 results
Python	3.12.13
TensorFlow	2.16.2
Device	CPU
Host	Turing
OS	macOS-26.4-arm64-arm-64bit
Machine	Apple M5 Max MacBook Pro, 64 GB RAM, 2TB SSD
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